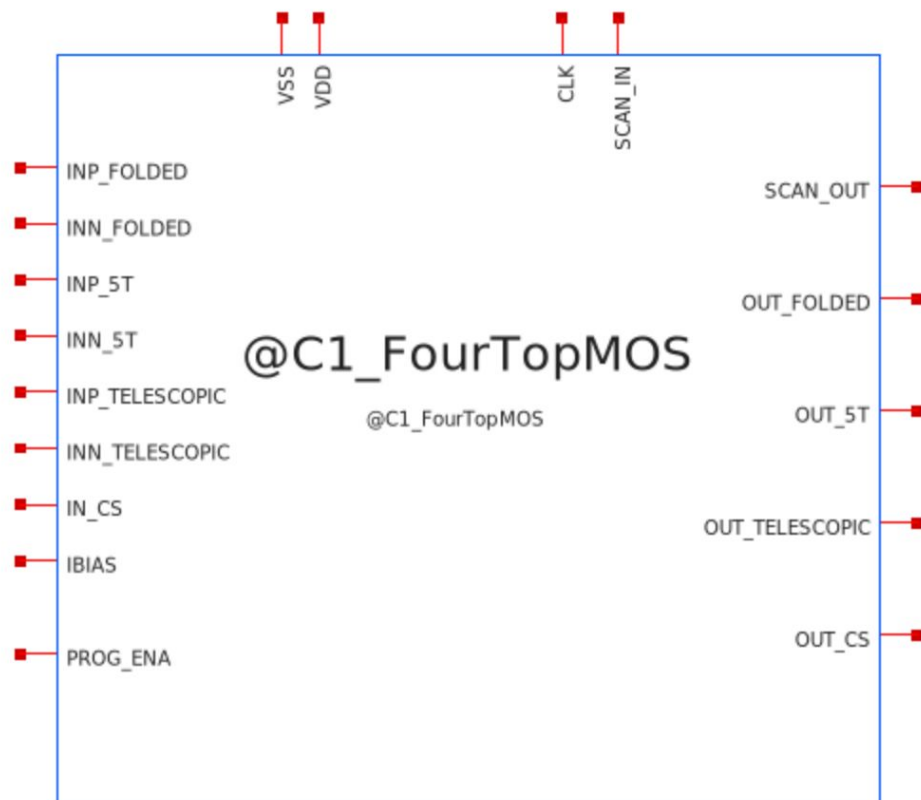


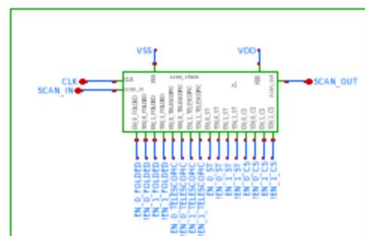
C1 Four Topology MOSbius

Brief PEX Results and Introduction

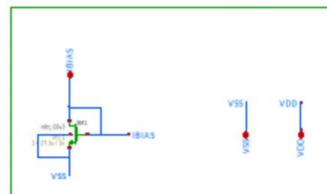
Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
boibun.	Columbia University	Undergraduate	Team Member
maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member



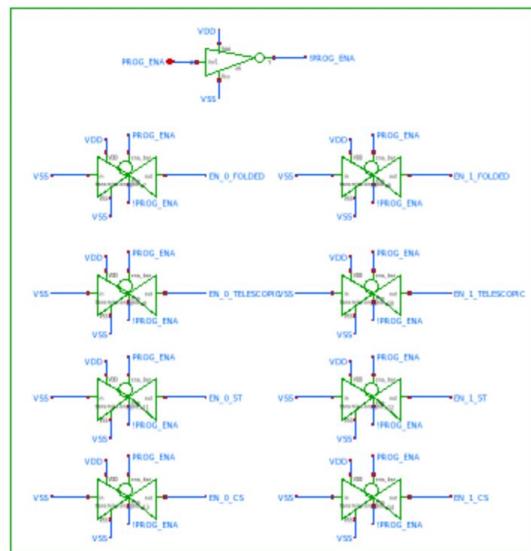
SCAN CHAIN



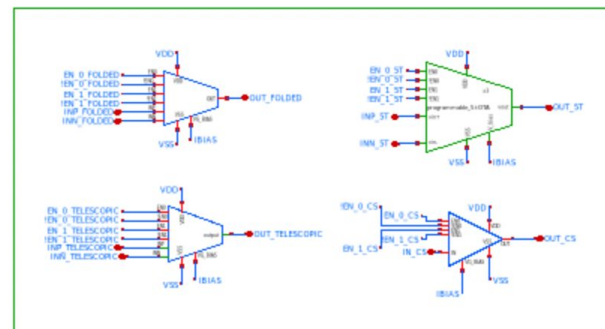
BIAS AND POWER PINS



PROGRAMMING ENABLE



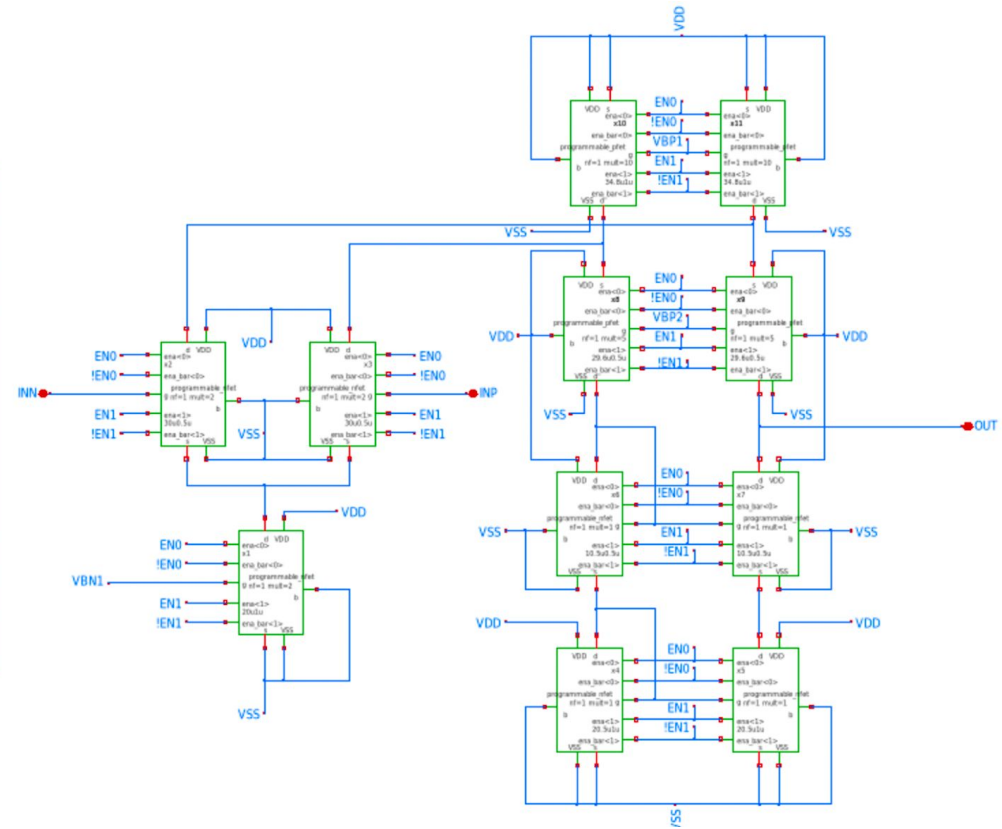
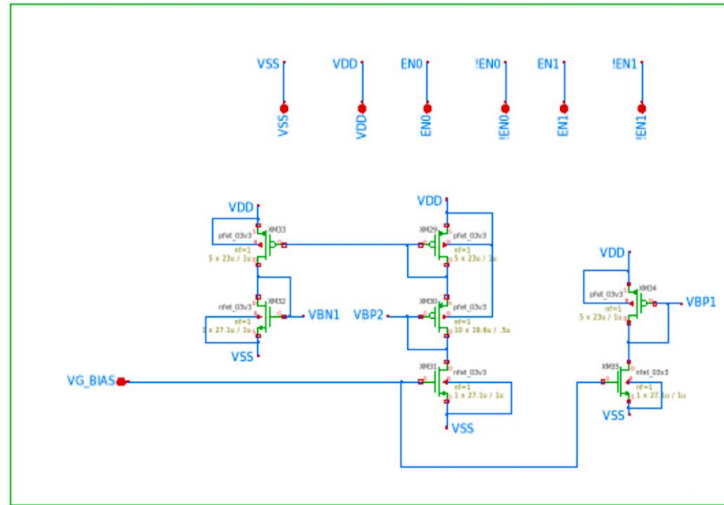
CORE AMPLIFIERS



Pull Downs on
Enable Pins



Bias Voltage Generation and Pins



Folded Cascode Sizing

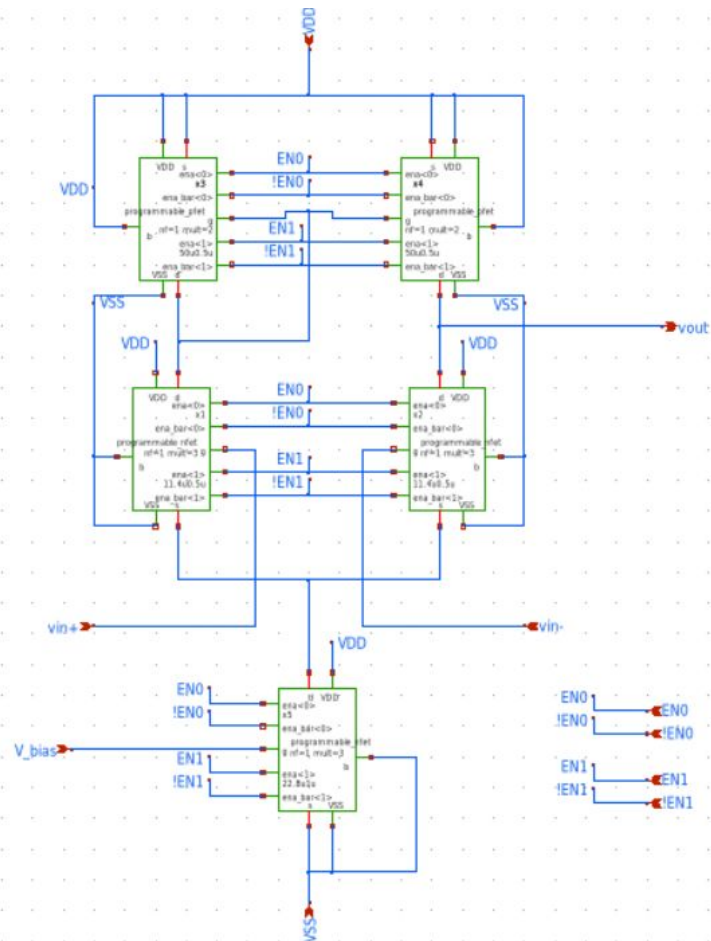
Device	Role	W (μm)	L (μm)	ID (μA)
XM10/XM11	NMOS input pair	60	0.5	75
XM12	NMOS tail	40	1	150
XM2/XM5	PMOS current sources	174	1	150
XM3/XM4	PMOS cascodes	148	0.5	75
XM6/XM7	NMOS cascodes	10.5	0.5	75
XM8/XM9	NMOS mirrors	20.5	1	75

(Default sizing shown, for 2x, 3x, and 4x sizing, the width and bias current would scale respectively for all amplifiers)



Telescopic Cascode Sizing

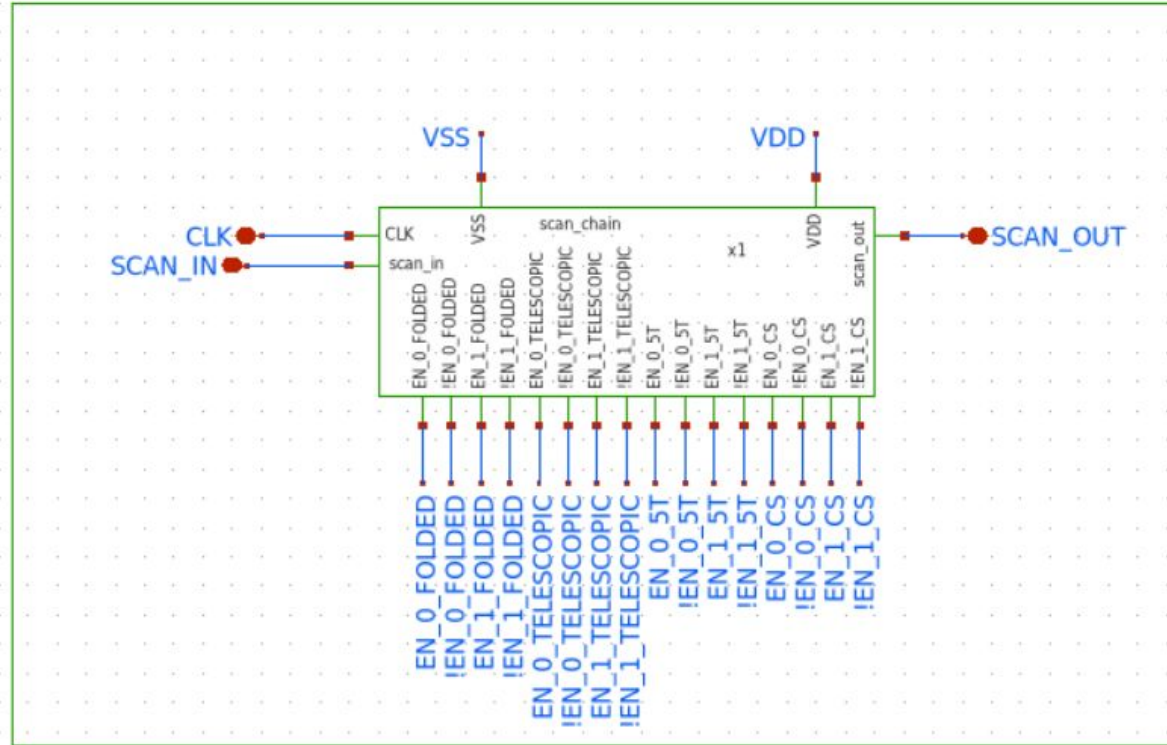
Device	Role	W (μm)	L (μm)	ID (μA)
XM1	NMOS Tail	60	0.5	200
XM2/XM3	Input Pair	30	0.5	100
XM4/XM5	NMOS Cascode	180	0.5	100
XM6/XM7	PMOS Cascode	235	0.4	100
XM8/XM9	PMOS Current Sources	240	0.5	100



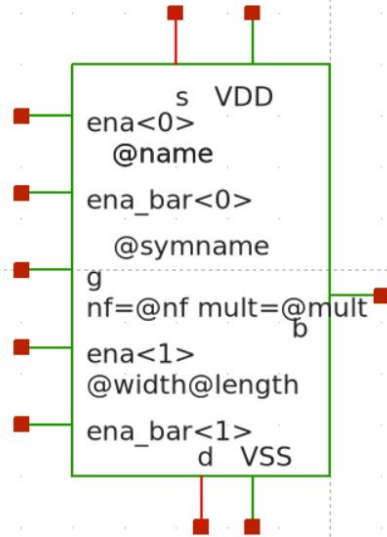
5-Transistor Sizing

Device	Role	W (μm)	L (μm)	ID (μA)
XM1/XM2	Input Pair	34.2	0.5	150
XM3/XM4	PMOS Active Load	100	0.5	150
XM5	NMOS Tail	68.4	0.5	300

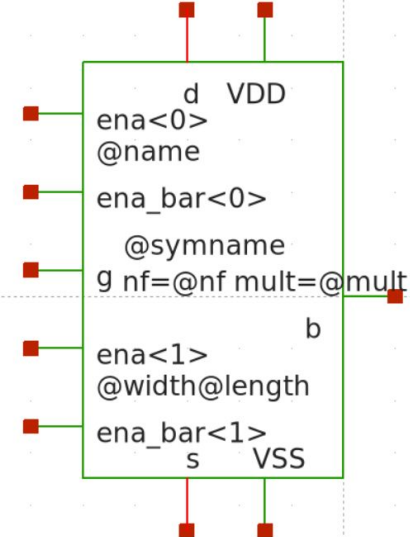
SCAN CHAIN



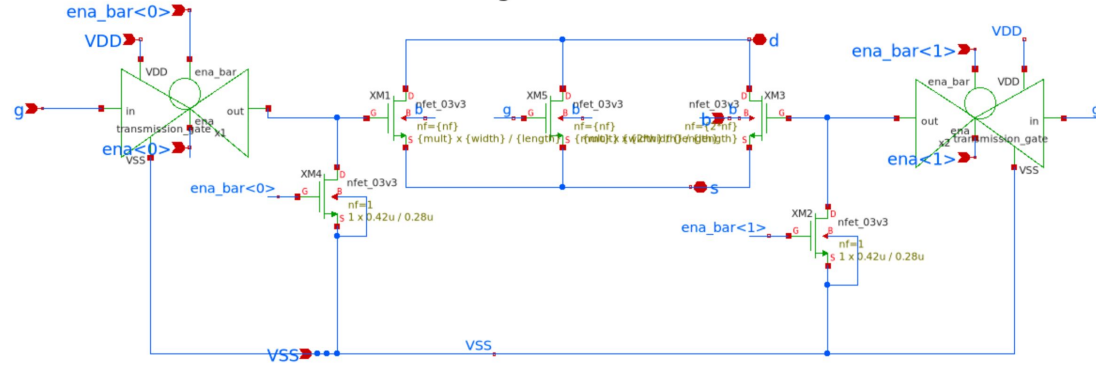
Programmable PFET



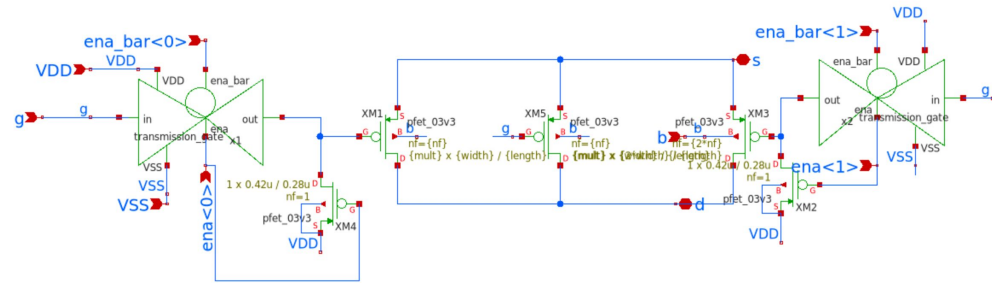
Programmable NFET



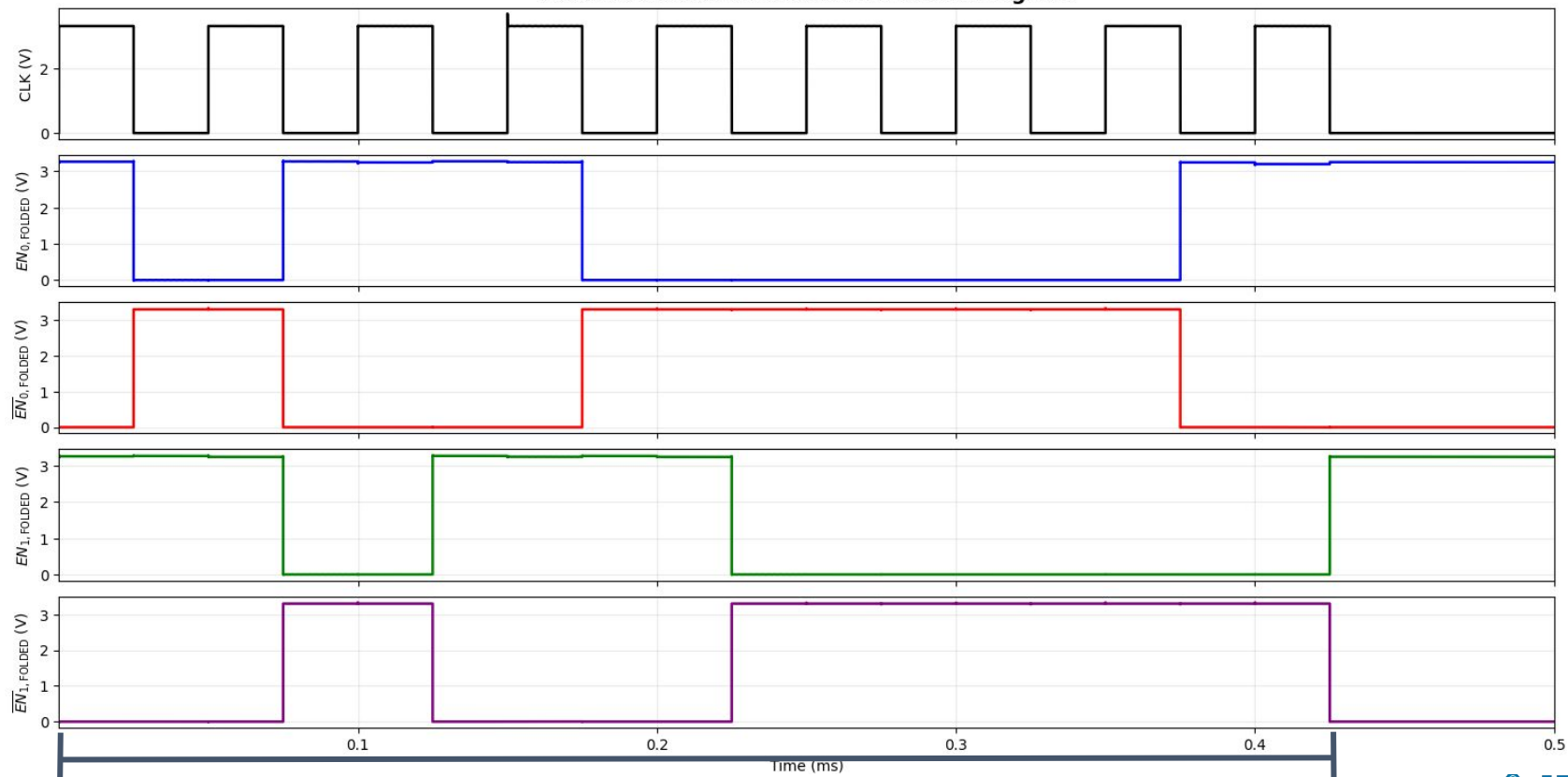
Programmable NFET



Programmable PFET

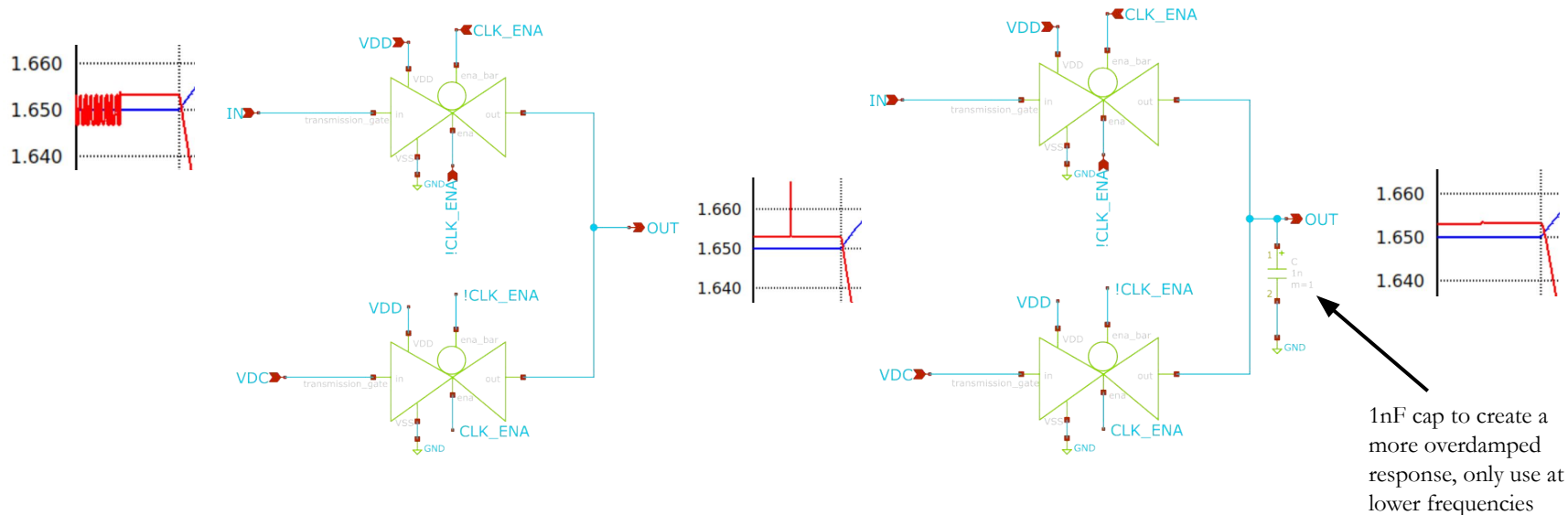


Folded Cascode OTA Transient Control Signals



.425 ms Data Propagation

Start-Up Cleaning Circuitry (SIMULATION ONLY)



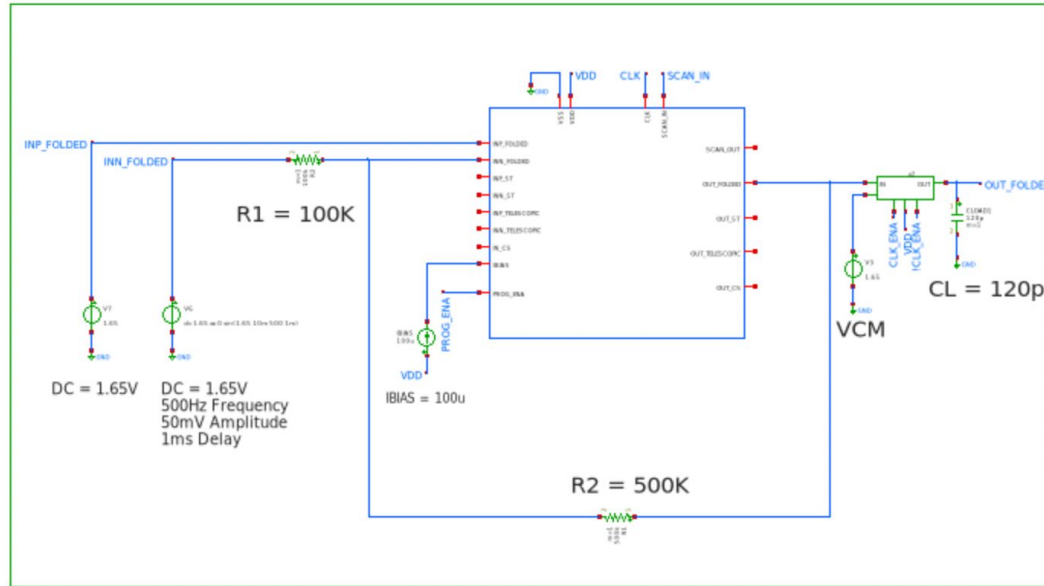
Start-up transients are present at the output when enable signals propagate through the scan chain. To mitigate this, a switch ties the output to VCM while the enable signal propagates. Once the enable logic stabilizes, the switch restores the actual output to the output pin

The following configuration uses the Four-Topology MOSbius set up as an inverting amplifier. The gain of 5 is established using external feedback resistors, $500\text{k}\Omega$ and $100\text{k}\Omega$. A switch is included to tie the outputs to VCM during start-up, preventing instability or undefined states as the circuit powers on. There is no additional capacitance present in this implementation of the switch. This setup also uses a PEX (parasitic-extracted) netlist.

Example Testbench

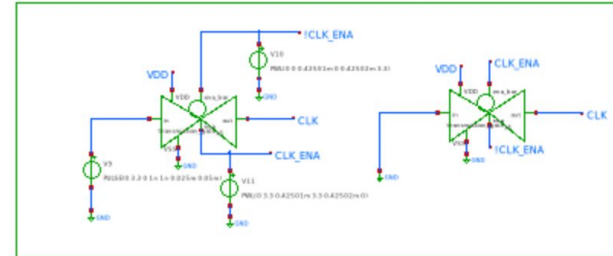
IMPLEMENTATION

Configure the folded cascode in 4x mode as an inverting amplifier to amplify a sine wave input by 5 while driving a 120pF capacitive load.



CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

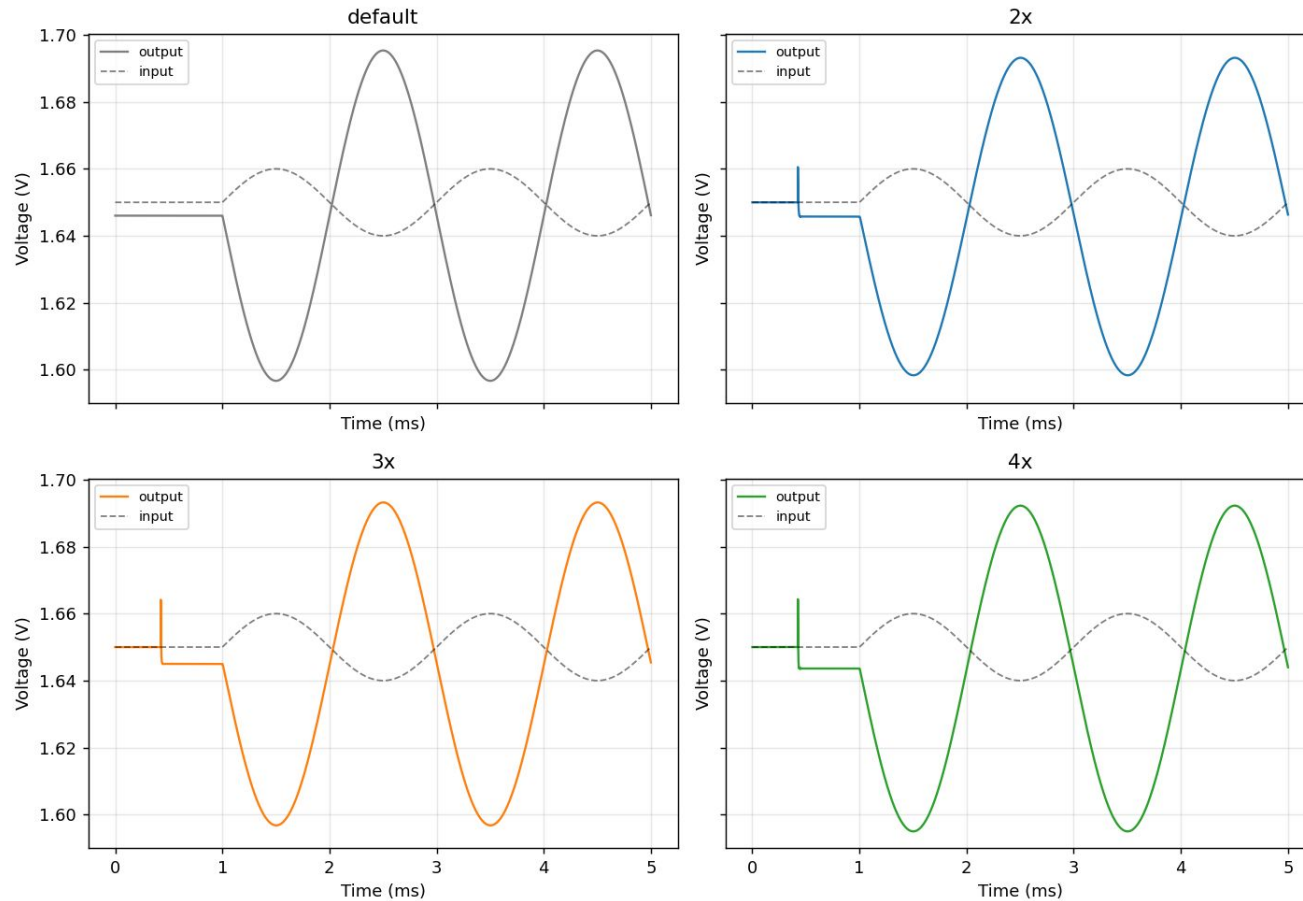


VDD, GND, SCAN INPUT, PROGRAMMABLE ENABLE

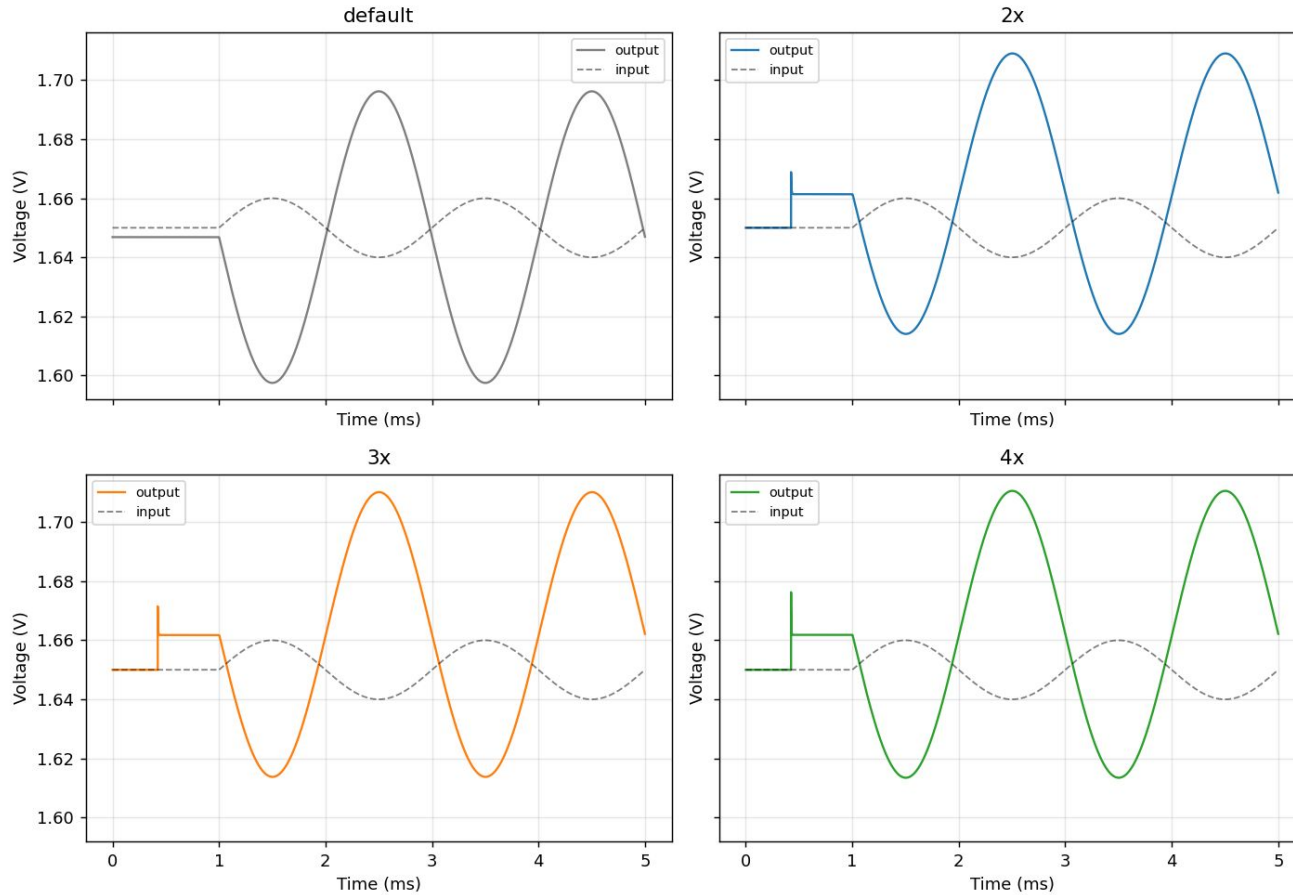
Scan chain input artificially generated to enable folded cascode 4x sizing
prog_ena sets default (low) or program (high)



Folded OTA Inverting Amplifier Simulation



Telescopic OTA Inverting Amplifier Simulation



5T OTA Inverting Amplifier Simulation

